

Gauss-law protection and exact fourth-order lifting of a T_1^{+-} glueball band in strongly coupled $SU(3)$ lattice gauge theory: compact localized states, a real-space sum of squares, global band edges, and universal $SU(N)$ second-order flatness

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Abstract

We determine the one-excitation band structure of strongly coupled $SU(3)$ Hamiltonian lattice gauge theory in 3+1 dimensions in exact rational arithmetic, and resolve the first order at which the lowest charge-conjugation-odd band becomes mobile. At second order the lowest C -odd branch is exactly flat over the Brillouin zone. Its flat subspace is $\ker B(k)^\dagger$, where the signed hopping symbol factors as $\tilde{N}(k) + 4I = B(k)B(k)^\dagger$; compact localized states are the consistently oriented boundaries of elementary cubes, and flatness is the chain-complex identity $\partial_2\partial_3 = 0$. All third-order tromino processes vanish by a bare-link lemma, so the band remains rigid through $O(y^3)$:

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 + O(y^4).$$

Using the complete Hermitian 189-record fourth-order real-space kernel, we show that this protection fails at the next order in a fully controlled way. For $X_i = 1 - \cos k_i$ and the cell-gauge flat vector $\psi = (e^{ik_2} - 1, -(e^{ik_1} - 1), e^{ik_0} - 1)^T$, the projected coefficient factorizes exactly as

$$\psi^\dagger [H_4(k) - qI] \psi = \frac{5}{12} \sum_i X_i^2 + \frac{17607806155349}{275331901291200} \sum_{i < j} X_i X_j,$$

where $\|\psi\|^2 = 2 \sum_i X_i$ and

$$q = -\frac{20721577909065127111}{7250590288602460800}.$$

This momentum-space factorization lifts exactly to a local real-space sum-of-squares theorem. If $C = \partial_3$ is the oriented cube-boundary map, T_i translates cube amplitudes, and $L_i = (T_i - I)^\dagger(T_i - I)$, then

$$C^\dagger (H_4 - qI) C = \frac{5}{48} \sum_i L_i^2 + \frac{17607806155349}{1101327605164800} \sum_{i < j} (\nabla_i \nabla_j)^\dagger (\nabla_i \nabla_j).$$

Thus the fourth-order mobility is carried by axial three-cube second differences and planar four-cube checkerboard differences. Positivity proves that Γ is the unique global minimum; elementary cube inequalities prove that $R = (\pi, \pi, \pi)$ is the unique global maximum. The

*Independent Researcher. Draft v0.5. All displayed strong-coupling constants and exact enumeration counts are machine-certified; see Appendix A.

fourth-order bandwidth is $\Delta W_4 \approx 0.4806178691$, with its exact rational value given below. The lift near Γ is quadratic but direction-dependent, so no ordinary effective-mass Hessian exists there; near R the curvature is isotropic and negative. Thus the Gauss-law band is exactly immobile through third order and acquires its first nonzero, parameter-free dispersion at fourth order. Independently, the shared-link calculus extends to every $SU(N)$ with integer $N \geq 3$: in the reduced normalization fixed by the $SU(3)$ domino,

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(4N^2 - 9)(2N^2 - 1)} > 0, \quad t_N \sim \frac{1}{4N^3},$$

and the group-independent incidence kernel gives a universally flat closed-surface branch at second order. At fourth order the complete $SU(3)$ source chain is now recovered from Stage 0 through the folded des-Cloizeaux Stage 3I and final kernel assembly Stage 3J. For every integer $N \geq 7$, the maximum local tensor degree six forces exact fundamental–antifundamental balance, eliminating all determinant/epsilon sectors. An exact stable-rank certificate reduces the full local problem to the $(1, 1)$, $(2, 2)$, and $(3, 3)$ Haar families, 140 balanced token signatures, and 94 nonvanishing denominator polynomials. Thus the stable-rank geometry and denominator layers are complete; the remaining load-bearing step is the symbolic walled-Brauer contraction producing q_N, A_N, B_N . The C -even band and the T_1^{+-} quantum-number assignment are also given exactly. No continuum or all-orders claim is made.

1 Introduction

Flat bands — single-particle dispersions that are exactly constant over the Brillouin zone — arise from destructive interference encoded in lattice geometry and are spanned by compact localized states (CLS) [7, 8, 9, 10, 11, 12]. They are usually discussed in engineered condensed-matter, photonic, and cold-atom systems. Here the same structure appears without geometric engineering in strongly coupled $SU(3)$ Hamiltonian lattice gauge theory [1].

In the strong-coupling regime the elementary one-flux excitations are single plaquettes of chromoelectric flux. Degenerate perturbation theory turns the three plaquette orientations into a 3×3 Bloch problem. The main results are:

- (A) At second order the lowest C -odd band is exactly flat, with coefficient $11/306$ at every momentum (Theorem 5.3).
- (B) The flat subspace is a lattice Gauss-law kernel. The signed adjacency factors through the plaquette-boundary incidence map, and the compact localized states are oriented boundaries of elementary cubes (Theorems 5.1 and 5.4).
- (C) Every third-order tromino amplitude vanishes by a bare-link argument. The flat band therefore shifts rigidly by the exact coefficient $-109151/249696$ (Theorem 6.2).
- (D) The complete fourth-order kernel breaks the flatness. Its projection onto the lower-order flat branch factorizes into two positive cubic invariants in $X_i = 1 - \cos k_i$ (Theorem 7.3).
- (E) The projected operator has an exact local real-space sum-of-squares representation. Its two elementary deformations are an axial three-cube second difference and a planar four-cube checkerboard difference (Theorem 7.4).
- (F) The factorization gives an analytic global theorem: Γ is the unique band minimum, R the unique band maximum, and the exact fourth-order bandwidth is $\Delta W_4 = A + B$ (Theorem 7.6).

- (G) The rest-point lift is direction-dependent. The Γ band coefficient is continuous and differentiable but has no ordinary Hessian; the R curvature is isotropic and negative (Proposition 7.8).
- (H) The rest-frame C -odd triplet transforms as T_1^{+-} , while the C -even sector splits as $A_1^{++} \oplus E^{++}$ and disperses normally (Theorems 5.3 and 5.5).
- (I) The signed shared-link hopping is positive for every $SU(N)$, $N \geq 3$, with exact rational coefficient t_N and large- N behavior $t_N \sim 1/(4N^3)$. Since the incidence kernel is group-independent, the closed-surface second-order branch is universally flat (Theorem 5.2).
- (J) At fourth order, every $SU(N)$ with $N \geq 7$ lies in a stable-rank regime: local degree is at most six, determinant/epsilon sectors disappear, and the complete geometry, balanced Haar families, representation channels, and electric denominators reduce to exact rational functions of N (Theorem 7.9).

The physical statement is therefore finite-order and sharp. A closed-surface T_1^{+-} excitation cannot propagate through the link-mediated channels that occur through $O(y^3)$: every boundary-link amplitude cancels exactly. At $O(y^4)$, site-mediated geometries activate and the cancellation no longer protects the projected branch. The resulting mobility is not inferred from a scan; it is fixed by an exact positive rational factorization.

The second-order calculation also corrects the C -even nearest-neighbor hop quoted in the companion manuscript [14]. The missing vacuum-mediated route changes it to $t_+ = -481/612 + 3/4 = -11/306$, independently adjudicated by exact two-plaquette diagonalization. The corresponding per-bond ratio $|t_-|/|t_+| = 5/22$ does not itself explain immobility; the incidence kernel does.

Strong-coupling expansions of Hamiltonian lattice gauge theory have a long history [1, 2, 3, 4], and the continuum glueball spectrum is accurately known from Euclidean Monte Carlo [5, 6]. The present statements are exact order-by-order results in the strong-coupling Hamiltonian expansion. They do not predict a continuum mass, do not establish convergence of the series, and do not claim all-orders flatness.

Section 2 fixes conventions. Sections 3 and 4 derive the one-plaquette and shared-link constants. Section 5 proves the second-order band structure and compact localized states. Section 6 proves third-order rigidity. Section 7 gives the link-channel robustness theorem, solves the $SU(3)$ fourth-order band analytically, derives its exact real-space SOS operator, and proves the stable-rank $SU(N \geq 7)$ geometry and denominator reduction. Section 8 summarizes the exact data and physical consequences; Section 9 lists the remaining problems.

2 Setting

We work with the Kogut–Susskind Hamiltonian [1] for $SU(3)$ on a cubic spatial lattice ($d = 3$),

$$H_\beta = \frac{1}{2} \sum_{\ell} C_2(\ell) + \beta \sum_p \left(1 - \frac{1}{N} \text{Re Tr } U_p \right), \quad N = 3, \quad (1)$$

in units of the electric coupling; $C_2(\ell)$ is the quadratic Casimir on link ℓ and U_p the oriented plaquette holonomy. We use the natural strong-coupling variable

$$y = \frac{2\beta}{3}, \quad \beta = \frac{3y}{2}, \quad (2)$$

in which the one-plaquette energy scale is $8/3$ and the perturbation per plaquette carries one power of y . The strong-coupling vacuum is the electric ground state $|0\rangle$ (all links in the trivial representation); the elementary excitations are single-plaquette flux states. With $\chi_p = \text{Tr } U_p$ the fundamental character on plaquette p , the normalized one-flux states of definite charge conjugation are

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad C|p, \pm\rangle = \pm|p, \pm\rangle, \quad (3)$$

with unperturbed energy $8/3$ (four links in the fundamental representation: $4 \times \frac{1}{2} \cdot \frac{4}{3} = \frac{8}{3}$; we keep the conventions of [14] throughout). On the cubic lattice plaquettes carry three orientations $(\mu\nu) \in \{xy, yz, zx\}$, so each momentum sector is a 3×3 problem.

Degenerate strong-coupling perturbation theory in the one-flux manifold produces, order by order in y , an effective hopping Hamiltonian $H_{\text{eff}}(k)$ per charge sector. Two structural facts organize the computation: (i) within-plaquette processes renormalize the diagonal and are captured exactly by the one-plaquette towers of Section 3; (ii) the leading inter-plaquette processes are mediated by *shared links* — each plaquette has exactly $4(2d-3) = 12$ shared-link neighbors in $d=3$ (four coplanar, eight transverse) — and reduce to balanced Haar moments on the shared link [14]. We use the linked-cluster bookkeeping of [14] (its Assumption 6.5); all finite-system counting is verified non-perturbatively by exact diagonalization on the two-plaquette “domino” (Section 4).

3 One-plaquette towers and the bridge

The $SU(3)$ one-plaquette class Hamiltonian $h_\beta = \frac{1}{2}C_2 + \beta(1 - \frac{1}{3}\text{Re Tr } g)$ restricted to class functions has charge-even and charge-odd gaps $\Delta_\pm(\beta)$ with exact strong-coupling towers [14]

$$\Delta_\pm(\beta) = \frac{2}{3} \mp \frac{\beta}{6} + b_2^\pm \beta^2 + b_3^\pm \beta^3 + O(\beta^4), \quad b_2^+ = \frac{13}{180}, \quad b_3^+ = \frac{101}{2700}, \quad b_2^- = \frac{1}{18}, \quad b_3^- = \frac{7}{432}. \quad (4)$$

The bridge to the lattice one-flux sector is the substitution $\beta = 3y/2$ with an overall factor 4: the *within-plaquette* part of the effective one-flux energy is $4\Delta_\pm(3y/2)$, i.e.

$$\underbrace{\frac{8}{3} \mp y + \frac{13}{20}y^2 + \frac{101}{200}y^3}_{C\text{-even}}, \quad \underbrace{\frac{8}{3} + y + \frac{1}{2}y^2 + \frac{7}{32}y^3}_{C\text{-odd}}, \quad (5)$$

to the orders needed here. Everything beyond (5) is inter-plaquette leakage, computed in the next two sections.

4 Second order: exact leakage and the corrected C -even hop

At $O(y^2)$ the only inter-plaquette geometry is the shared-link pair. Integrating the six free links of the pair by the fusion identity $\int dB \text{Tr}(XB) \text{Tr}(YB^\dagger) = \text{Tr}(XY^\dagger)/3$ reduces every required quantity to balanced $(2, 2)$ Haar moments on the shared link, resolved into the irreducible channels $\{1, 8, \bar{3}, 6\}$ of the fused link representation, with dimension-weighted norms $N_R = d_R/9$. The corresponding intermediate energies are $\{4, \frac{11}{2}, \frac{14}{3}, \frac{17}{3}\}$, to be used in the denominators $E_0 - E_R$ with $E_0 = 8/3$ [14].

Lemma 4.1 (Channel sums). *Per shared-link neighbor, the fused-channel second-order sum is*

$$\Sigma_{\text{chan}} = -\frac{1}{12} - \frac{16}{51} - \frac{1}{6} - \frac{2}{9} = -\frac{481}{612}, \quad (6)$$

independent of the relative link orientation, and equal for the self-energy and hopping networks (the operator identity $\langle G_{p'}|W_p = \langle G_p|W_{p'}$ of [14]). The diagonal carries in addition the vacuum subtraction $+3/4$ per neighbor.

The companion manuscript identified Σ_{chan} with the full hop. It is not: the hop has one more intermediate state.

Lemma 4.2 (Vacuum-mediated route). *For C -even one-flux states the second-order effective Hamiltonian contains the vacuum-intermediate route*

$$\langle e_r|W R W|e_i\rangle \supset \frac{\langle e_r|W|0\rangle\langle 0|W|e_i\rangle}{8/3-0} = (\sqrt{2})(\sqrt{2})\frac{3}{8} = +\frac{3}{4} \quad (7)$$

for every ordered pair (i, r) , adjacent or distant. It is absent in the C -odd sector ($\langle o|W|0\rangle = 0$ by parity). For distant pairs it exactly cancels the free two-flux channel sum $2/(8/3 - 16/3) = -3/4$, recovering the vanishing of distant hops; for shared-link neighbors the cancellation against Lemma 4.1 is partial.

Theorem 4.3 (Second-order leakage, both sectors). *Per shared-link neighbor:*

$$C\text{-even: self-energy} = -\frac{481}{612}, \quad \text{vacuum subtraction} = +\frac{3}{4}, \quad t_+ = -\frac{481}{612} + \frac{3}{4} = -\frac{11}{306}; \quad (8)$$

$$C\text{-odd: self-energy} = -\frac{481}{612} \text{ (} C\text{-independent)}, \quad t_-(s) = \frac{5s}{612}, \quad (9)$$

where $s = \pm 1$ is the relative orientation of the shared link in the two plaquette boundaries (the incidence product), and the C -odd hop has no vacuum route. The C -odd channel-family split is $N_{\text{mixed}} = -\frac{27}{68}$ (channels 1, 8), $N_{\text{like}} = -\frac{7}{18}$ (channels $\bar{3}$, 6), with sum $-\frac{481}{612}$ and difference $\frac{5}{612}$.

Remark 4.4 (Adjudication). Equation (8) corrects eq. (22) of [14] ($-9397/1020 \rightarrow -217/1020$ after assembly; see Theorem 5.5). The correction is adjudicated by the companion's own exact rational diagonalization of (1) on the two-plaquette domino, performed blind to the perturbative assembly: the C -even one-flux levels are $\{\frac{1769}{3060}, \frac{13}{20}\}$ at the relevant order, i.e. $\text{diag} \pm |t_+|$ with $\text{diag} = \frac{1879}{3060}$, forcing $|t_+| = \frac{110}{3060} = \frac{11}{306}$; the C -odd pair $\{\frac{31}{68}, \frac{17}{36}\}$ gives mean $\frac{71}{153} = \frac{1}{2} - \frac{11}{306}$ and half-splitting exactly $\frac{5}{612}$. Every constant in this section is additionally certified by a 251-gate exact-arithmetic computer proof (Appendix A).

5 The exact band structure at second order

Assemble the one-flux sector at momentum k : a 3×3 Bloch problem over plaquette orientations. Let $A(k)$ and $\tilde{N}(k)$ denote the unsigned and signed Bloch matrices of the 12-regular shared-link adjacency (signs $s = \sigma_p(b)\sigma_{p'}(b)$, the incidence product on the shared link). Then through $O(y^2)$

$$H_{\text{eff}}^{(+)}(k) = \left[\frac{8}{3} - y + \frac{13}{20}y^2\right]I + y^2\left[12\left(-\frac{481}{612} + \frac{3}{4}\right)I + t_+A(k)\right] = \left[\frac{8}{3} - y\right]I + y^2\left[\frac{223}{1020}I - \frac{11}{306}A(k)\right], \quad (10)$$

$$H_{\text{eff}}^{(-)}(k) = \left[\frac{8}{3} + y\right]I + y^2\left[\frac{7}{102}I + \frac{5}{612}\tilde{N}(k)\right]. \quad (11)$$

Theorem 5.1 (Incidence factorization; Gauss law). *Let $N(k)$ and $B(k)$ be the unsigned and signed 3×3 Bloch incidence symbols of the oriented plaquette-boundary map (each plaquette feeds its four boundary links with incidence signs). With $u_j = 1 - e^{ik_j}$, $v_j = 1 + e^{ik_j}$,*

$$A(k) + 4I = N(k)N(k)^\dagger, \quad \det N(k) = -2v_1v_2v_3; \quad \tilde{N}(k) + 4I = B(k)B(k)^\dagger, \quad \det B(k) \equiv 0. \quad (12)$$

Hence $\lambda(k) \in [-4, 12]$ and $\mu(k) \in [-4, 8]$ for the spectra of A and $\tilde{N}$, and the C -odd spectrum contains $\mu = -4$ at every k : the lowest C -odd band is the kernel of $B(k)^\dagger$, the states with zero net signed amplitude into every link — a lattice Gauss law.

Proof. B is the boundary symbol: row $(\mu\nu)$ has entries $e^{-ik_\nu/2} - e^{ik_\nu/2}$ and $e^{ik_\mu/2} - e^{-ik_\mu/2}$ in the link channels μ, ν (plaquette-center gauge) and 0 in the transverse channel; $BB^\dagger$ reproduces, entry by entry, the signed adjacency plus $4I$ (each plaquette has four boundary links), which is verified symbolically. Each row of B is a linear combination of two link channels, and the three rows involve the three channels cyclically with proportional weights; the 3×3 determinant cancels identically (a two-term cancellation $\propto \sin \frac{k_0}{2} \sin \frac{k_1}{2} \sin \frac{k_2}{2}$). The unsigned statement is the same computation without signs. $\square$

Theorem 5.2 ($SU(N)$ shared-link universality). *For every integer $N \geq 3$, let F denote the fundamental representation, $C_F = (N^2 - 1)/(2N)$, and use the reduced shared-link normalization in which the $SU(3)$ signed hop is $5/612$. The mixed-orientation fusion $F \otimes \bar{F} = 1 \oplus \text{Adj}$ and the like-orientation fusion $F \otimes F = A_2 \oplus S_2$ give*

$$\begin{aligned} w_1 &= -\frac{2}{N(N-1)(N+1)}, & w_{\text{Adj}} &= -\frac{2(N-1)(N+1)}{N(2N^2-1)}, \\ w_{A_2} &= -\frac{N-1}{(N+1)(2N-3)}, & w_{S_2} &= -\frac{N+1}{(N-1)(2N+3)}. \end{aligned} \quad (13)$$

Consequently

$$\begin{aligned} W_{\text{mixed}} &= -\frac{2N^3}{(N^2-1)(2N^2-1)}, \\ W_{\text{like}} &= -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)}, \\ \Sigma_N &:= W_{\text{like}} + W_{\text{mixed}} = -\frac{2N(8N^4-19N^2+4)}{(N^2-1)(4N^2-9)(2N^2-1)}, \\ t_N &:= W_{\text{like}} - W_{\text{mixed}} = \frac{2N(N^2-4)}{(N^2-1)(4N^2-9)(2N^2-1)} > 0. \end{aligned} \quad (14)$$

The second-order C -odd operator therefore has the form $H_{2,N}(k) = d_N I + t_N \tilde{N}(k)$ for a scalar d_N . Since $\tilde{N}(k)$ has the exact eigenvalue -4 at every momentum, every $SU(N)$, $N \geq 3$, has a momentum-independent closed-surface eigenvalue $d_N - 4t_N$ at second order. Moreover,

$$\Sigma_3 = -\frac{481}{612}, \quad t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} + O(N^{-7}).$$

Proof. After the six free-link integrations, the normalized shared-link tensor is isotropic on the N^2 -dimensional product space. If P_R is the orthogonal projector onto an irreducible channel R , its squared channel norm is $\text{Tr } P_R / N^2 = d_R / N^2$. The six nonshared fundamental links contribute $3C_F$ to the intermediate electric energy, while the fused shared link contributes $C_R/2$; hence

$$E_0 = 2C_F, \quad E_R = 3C_F + \frac{C_R}{2}, \quad w_R = -\frac{d_R/N^2}{C_F + C_R/2}.$$

Using

$$d_1 = 1, \quad C_1 = 0; \quad d_{\text{Adj}} = N^2 - 1, \quad C_{\text{Adj}} = N;$$

$$d_{A_2} = \frac{N(N-1)}{2}, \quad C_{A_2} = \frac{(N+1)(N-2)}{N}; \quad d_{S_2} = \frac{N(N+1)}{2}, \quad C_{S_2} = \frac{(N-1)(N+2)}{N}$$

gives (13). Summing the two orientation families and subtracting them gives (14). Every displayed factor in its numerator and denominator is positive for $N \geq 3$. The flat eigenvalue then follows from Theorem 5.1; the asymptotic expansion is direct. $\square$

Theorem 5.3 (Exactly flat T_1^{+-} band; compact localized states). *The lowest C -odd band is exactly flat:*

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306} y^2 + O(y^3) \quad \text{for every } k \in \text{BZ}, \quad (15)$$

since $\frac{7}{102} + \frac{5}{612}(-4) = \frac{11}{306}$. The dispersive C -odd partners span coefficients $[\frac{11}{306}, \frac{41}{306}]$, touching the flat band quadratically at $k = 0$. At rest the C -odd triplet transforms as T_1^{+-} of the cubic group with $C = -$ (axial-vector-like): the C -odd one-plaquette sector contains no scalar at rest.

Theorem 5.4 (CLS, completeness, and $\partial\partial = 0$). *The consistently (Levi-Civita) oriented boundary of a single elementary cube — six faces with amplitudes ± 1 — is an exact real-space eigenvector of the signed adjacency at $\mu = -4$ with zero leakage onto any plaquette off the cube. Its Bloch symbol $u(k)$ has $|u(k)|^2 = \sum_a \sin^2(k_a/2)$ and spans $\ker B(k)^\dagger$ for all $k \neq 0$. Flatness is the chain-complex identity $\partial_2\partial_3 = 0$: every edge of the cube is shared by exactly two faces with opposite induced orientation, so the signed face amplitudes cancel on all twelve edges. Exactly, on an L^3 torus,*

$$\dim \ker(\tilde{N} + 4I) = L^3 + 2 = \underbrace{(L^3 \text{ cube states, rank } L^3 - 1)}_{\text{one global relation } \sum_{\text{cubes}} \psi = 0} \oplus \underbrace{(3 \text{ rest states})}_{\tilde{N}(0) = -4I}. \quad (16)$$

Theorem 5.5 (The C -even band). *With the corrected hop (8),*

$$E_+(k, y) = \frac{8}{3} - y + y^2 \left[\frac{223}{1020} - \frac{11}{306} \lambda(k) \right], \quad \lambda(k) \in [-4, 12], \quad (17)$$

with band bottom $-\frac{217}{1020}$ at $k = 0$ (the $A_1^{++}, 0^{++}$ state), where $\lambda(k) = 12 - \frac{4}{3}|k|^2 + O(k^4)$ isotropically, hence curvature $+\frac{22}{459}|k|^2 y^2$ and effective mass $m^* = 459/(44 y^2)$; band top $\frac{1109}{3060}$ at the zone faces; bandwidth $16|t_+| = \frac{88}{153}$. At rest the triplet decomposes as $A_1^{++} \oplus E^{++}$, with the E^{++} doublet at the exact, hop-independent coefficient $\frac{223}{1020}$ (it sits at $\lambda = 0$).

Remark 5.6 (An exactly immobile excitation). *The per-bond ratio is*

$$\frac{|t_-|}{|t_+|} = \frac{5/612}{11/306} = \frac{5}{22},$$

and carries no immobility claim (this corrects Remark 6.4 of [14], whose ratio $5/481$ used the uncorrected hop). The immobility statement holds in exact form: the lowest C -odd branch has bandwidth zero, while the full C -odd manifold has width $\frac{5}{51}$ against the C -even bandwidth $\frac{88}{153}$ (manifold-width ratio $15/88$). Both sectors factorize through the link channel; the qualitative asymmetry is that $\det N = -2v_1 v_2 v_3$ vanishes only on the zone faces $k_j = \pi$, while $\det B \equiv 0$ everywhere — the C -even kernel is measure-zero, the C -odd kernel is a whole band.

6 Third order: tromino vanishing and a rigid flat band

At $O(y^3)$ the new lattice geometry is the tromino: three plaquettes pairwise sharing links.

Lemma 6.1 (Bare-link lemma). *Every $O(y^3)$ tromino amplitude vanishes: each candidate three-plaquette process leaves at least one boundary link in a nontrivial representation unmatched (“bare”), and the corresponding Haar integral over that link is zero. Trominoes first activate at $O(y^4)$.*

Consequently the third-order effective Hamiltonian retains the second-order (signed-adjacency, link-mediated) structure. The C -odd hopping is odd in the shared-link orientation,

$$T_3(s) = -T_3(-s),$$

and the per-neighbor constants are computable on the abstract domino.

Theorem 6.2 (Rigid flat band at third order). *The flat band of Theorem 5.3 is rigid at $O(y^3)$:*

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306} y^2 - \frac{109151}{249696} y^3 + O(y^4) \quad \text{for every } k, \quad (18)$$

with

$$d_3 = \frac{7}{32} + 12 \text{leak}_3 - 4 b_3 = -\frac{109151}{249696}, \quad b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696}, \quad (19)$$

where $\frac{7}{32}$ is the within-plaquette tower term (5), b_3 the third-order C -odd hop and leak_3 the per-neighbor diagonal leakage; the dispersive C -odd top has cubic coefficient $-\frac{61751}{249696}$. In the C -even sector,

$$m_+(0, y) = \frac{8}{3} - y - \frac{217}{1020} y^2 - \frac{54049}{520200} y^3 + O(y^4), \quad (20)$$

the band-top cubic coefficient is $\frac{471353}{1560600}$, and the diagonal and hopping corrections obey the exact identity $\text{leak}_n^e = T_n^e$ at $n = 2, 3$ (two distinct “+3/4” mechanisms of equal value), so the A_1^{++} correction at $k = 0$ is (tower) $+24 T_n^e$ per order. From the certified band form, the E^{++} level acquires the cubic coefficient $\frac{101}{200} + 12 T_3^e = \frac{52163}{260100}$ and the C -even curvature the correction $+\frac{4}{3} |T_3^e| y^3 = \frac{6335}{187272} y^3$.

7 Robustness and exact fourth-order lifting

Theorem 7.1 (Link-mediated robustness). *Let $H_{\text{corr}}(k) = B(k)M(k)B(k)^\dagger$ be any correction whose symbol factors through the link channel, with $M(k)$ an arbitrary Hermitian symbol. Then H_{corr} annihilates the flat subspace:*

$$H_{\text{corr}}(k)u(k) = B(k)M(k)(B(k)^\dagger u(k)) = 0.$$

Consequently every link-mediated correction preserves the flat branch. A separate k -independent scalar multiple of the identity may shift its energy without producing dispersion. This explains the exact second- and third-order rigidity.

Theorem 7.2 (Sharpness). *The minimal corner-sharing, site-mediated symbol*

$$H_{\text{corner}}(k) = \text{diag}(2 \cos(k_0 + k_1), 2 \cos(k_1 + k_2), 2 \cos(k_0 + k_2))$$

does not preserve the lower-order flat subspace: $P_{\text{flat}} H_{\text{corner}} P_\perp \neq 0$. Geometry alone does not protect the band after site-mediated processes activate.

7.1 The complete fourth-order kernel

Let the exact translation-invariant fourth-order kernel be $K_4(r)_{ba}$, where $a, b \in \{01, 02, 12\}$ label plaquette orientations and $r \in \mathbb{Z}^3$ is the relative cell displacement. The recovered kernel contains 189 nonzero rational records and obeys exact real-space Hermiticity,

$$K_4(r)_{ba} = K_4(-r)_{ab}.$$

Its Bloch matrix is

$$H_4(k) = \sum_r K_4(r) e^{ik \cdot r}.$$

At zero momentum the full matrix is scalar,

$$H_4(0) = qI_3, \quad q = -\frac{20721577909065127111}{7250590288602460800}. \quad (21)$$

This is both an exact kernel sum and the cubic-symmetry requirement on the rest-frame T_1^{+-} triplet.

For $k \neq 0$ modulo reciprocal-lattice periodicity, use the cell/origin-gauge flat vector

$$\psi(k) = \begin{pmatrix} e^{ik_2} - 1 \\ -(e^{ik_1} - 1) \\ e^{ik_0} - 1 \end{pmatrix}, \quad \|\psi(k)\|^2 = 2 \sum_{i=0}^2 X_i, \quad X_i = 1 - \cos k_i. \quad (22)$$

It spans the one-dimensional lower-order flat eigenspace away from the threefold band touching at Γ .

Theorem 7.3 (Exact fourth-order factorization). *Define*

$$D(k) = \psi(k)^\dagger [H_4(k) - qI] \psi(k).$$

The complete 189-record kernel gives the identity

$$D(k) = A \sum_i X_i^2 + B \sum_{i < j} X_i X_j, \quad A = \frac{5}{12}, \quad B = \frac{17607806155349}{275331901291200}. \quad (23)$$

Therefore the fourth-order coefficient of the branch that is flat through third order is

$$c_4(k) = q + \frac{A \sum_i X_i^2 + B \sum_{i < j} X_i X_j}{2 \sum_i X_i}, \quad k \neq \Gamma, \quad (24)$$

with continuous value $c_4(\Gamma) = q$.

Proof. Contracting the exact Laurent-polynomial kernel with (22) leaves 25 nonzero frequencies. Hermiticity and cubic symmetry reduce them to three cosine orbits: unit-axis frequencies, double-axis frequencies, and the six pair frequencies. Thus

$$D = C + a \sum_i \cos k_i + d \sum_i \cos 2k_i + b \sum_{i < j} \{\cos(k_i + k_j) + \cos(k_i - k_j)\}.$$

The exact kernel coefficients satisfy

$$C + 3a + 3d + 6b = 0, \quad 2d = \frac{5}{12}, \quad 2b = B, \quad -a - 4d - 4b = 0.$$

The first identity is $D(0) = 0$; the last is the cancellation of the term linear in $X_i = 1 - \cos k_i$, equivalently the continuity of $D(k)/(2 \sum_i X_i)$ with value zero at Γ . Substitution of $\cos 2k_i = 1 - 4X_i + 2X_i^2$ and $\cos(k_i + k_j) + \cos(k_i - k_j) = 2(1 - X_i)(1 - X_j)$ gives (23). Equation (24) follows from $\|\psi\|^2 = 2 \sum_i X_i$. At Γ , (21) gives the continuous value directly. $\square$

The complete strong-coupling branch through fourth order is consequently

$$m_-(k, y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 + c_4(k)y^4 + O(y^5). \quad (25)$$

Theorem 7.4 (Exact real-space sum of squares). *Let $C = \partial_3$ map scalar cube amplitudes to their consistently oriented plaquette boundaries in the cell/origin gauge, so its Fourier symbol is $\psi(k)$ in (22). Let T_i translate cube amplitudes by one cell and define*

$$\nabla_i = T_i - I, \quad L_i = \nabla_i^\dagger \nabla_i = 2I - T_i - T_i^{-1}.$$

Then the complete fourth-order kernel obeys the exact local operator identity

$$\boxed{C^\dagger(H_4 - qI)C = \frac{A}{4} \sum_i L_i^2 + \frac{B}{4} \sum_{i < j} L_i L_j} \quad (26)$$

with A, B from (23). Equivalently,

$$C^\dagger(H_4 - qI)C = \frac{5}{48} \sum_i L_i^2 + \frac{17607806155349}{1101327605164800} \sum_{i < j} (\nabla_i \nabla_j)^\dagger (\nabla_i \nabla_j). \quad (27)$$

The scalar numerator operator $Q = C^\dagger(H_4 - qI)C$ is the 25-point stencil

$$\begin{aligned} (Q\phi)_x &= w_0 \phi_x + w_1 \sum_i (\phi_{x+e_i} + \phi_{x-e_i}) + w_2 \sum_i (\phi_{x+2e_i} + \phi_{x-2e_i}) \\ &\quad + w_d \sum_{i < j} \sum_{\sigma, \tau = \pm 1} \phi_{x+\sigma e_i + \tau e_j}, \end{aligned} \quad (28)$$

where

$$\begin{aligned} w_0 &= \frac{189690244462349}{91777300430400}, \\ w_1 &= -\frac{132329431693349}{275331901291200}, \\ w_2 &= \frac{5}{48}, \quad w_d = \frac{17607806155349}{1101327605164800}, \end{aligned} \quad (29)$$

and $w_0 + 6w_1 + 6w_2 + 12w_d = 0$.

The cube-boundary states have Gram operator

$$G = C^\dagger C = \sum_i L_i. \quad (30)$$

Consequently the physical lift is the generalized eigenproblem

$$Q\phi = \lambda G\phi, \quad \lambda(k) = c_4(k) - q = \frac{A \sum_i X_i^2 + B \sum_{i < j} X_i X_j}{2 \sum_i X_i}. \quad (31)$$

Proof. The symbol of L_i is $|e^{ik_i} - 1|^2 = 2X_i$. Hence the Fourier symbol of the right side of (26) is

$$\frac{A}{4} \sum_i (2X_i)^2 + \frac{B}{4} \sum_{i < j} (2X_i)(2X_j) = A \sum_i X_i^2 + B \sum_{i < j} X_i X_j,$$

which equals the exact contracted symbol $D(k)$ by Theorem 7.3. Since both sides are finite-range translation-invariant operators, equality of their Laurent symbols proves the real-space identity. Expanding L_i^2 and $L_i L_j$ gives (28)–(29). Equation (30) follows from the definition of C , and division by its symbol $2 \sum_i X_i$ gives (31). The exact certificate also contracts the 189-record kernel directly and obtains the same 25 rational coefficients with zero residual. $\square$

Remark 7.5 (Geometric carriers of fourth-order mobility). *For a localized cube amplitude, L_i creates the axial three-cube pattern $(-1, 2, -1)$, while $\nabla_i \nabla_j$ creates a planar four-cube checkerboard $(+1, -1, -1, +1)$. Equation (27) states that the first mobility of the closed-surface excitation is exactly the positive weighted norm of these two local deformations.*

Theorem 7.6 (Exact global band edges). *Modulo reciprocal-lattice periodicity, $\Gamma = (0, 0, 0)$ is the unique global minimum of $c_4(k)$ and $R = (\pi, \pi, \pi)$ is the unique global maximum. The exact fourth-order bandwidth is*

$$\Delta W_4 = c_4(R) - c_4(\Gamma) = A + B = \frac{132329431693349}{275331901291200}. \quad (32)$$

Proof. For $k \neq \Gamma$, the variables satisfy $X_i \in [0, 2]$ and $S = \sum_i X_i > 0$. Since $A > 0$, $B > 0$, and $\sum_i X_i^2 > 0$, equation (23) gives $D(k) > 0$. Hence $c_4(k) > q = c_4(\Gamma)$.

For the upper edge, write $Q = \sum_i X_i^2$ and $R_2 = \sum_{i < j} X_i X_j$. On the cube $0 \leq X_i \leq 2$,

$$Q \leq 2S, \quad R_2 \leq 2S.$$

For the second inequality set $x_i = X_i/2$; then

$$(x_0 + x_1 + x_2) - (x_0 x_1 + x_0 x_2 + x_1 x_2) = x_0(1 - x_1) + x_1(1 - x_2) + x_2(1 - x_0) \geq 0.$$

Therefore $c_4(k) - q \leq A + B$. Equality away from Γ requires $x_0 = x_1 = x_2 = 1$, hence $X_0 = X_1 = X_2 = 2$, which is precisely $R = (\pi, \pi, \pi)$ modulo 2π . $\square$

Proposition 7.7 (Kinematic generic- N factorization). *Fix an integer $N \geq 3$ and define*

$$D_N(k) = \psi(k)^\dagger [H_{4,N}(k) - q_N I] \psi(k).$$

Suppose the exact rank- N kernel passes the group-independent gates of Hermiticity, cubic covariance, independent-reflection symmetry, the same final displacement support as the $SU(3)$ kernel, and $H_{4,N}(0) = q_N I$. Then

$$D_N(k) = A_N \sum_i X_i^2 + B_N \sum_{i < j} X_i X_j.$$

Writing q_N, c_X, c_M, c_R for the exact fourth-order branch coefficients at Γ, X, M, R , respectively,

$$A_N = c_X - q_N, \quad B_N = 2(c_M - c_X) = c_R - c_X, \quad c_R = 2c_M - c_X.$$

The last equality is a hard consistency gate on any generic- N computation. This proposition does not assert that the symbolic contraction already passes the listed kernel gates, nor does it assert the signs of A_N, B_N beyond the certified $SU(3)$ case.

Proof. The final displacement support and cubic/reflection symmetries restrict D_N to a constant, a term proportional to $S = \sum_i X_i$, and the two quadratic cubic invariants $Q = \sum_i X_i^2$ and $R_2 = \sum_{i<j} X_i X_j$. Since $\psi(k) = O(k)$ and $H_{4,N}(k) - q_N I = O(k)$, analyticity gives $D_N(k) = O(|k|^3)$. Independent reflections make D_N even, hence in fact $D_N(k) = O(|k|^4)$. The constant and quadratic Taylor terms therefore vanish, leaving $A_N Q + B_N R_2$. Evaluating $(A_N Q + B_N R_2)/(2S)$ at X , M , and R gives A_N , $A_N + B_N/2$, and $A_N + B_N$, respectively, yielding the stated reconstruction identities. $\square$

The high-symmetry values are

$$\begin{aligned} c_4(\Gamma) &= -\frac{20721577909065127111}{7250590288602460800}, \\ c_4(X) &= -\frac{17700498622147435111}{7250590288602460800}, \\ c_4(M) &= -\frac{4367164159624988707}{1812647572150615200}, \\ c_4(R) &= -\frac{3447362930970494909}{1450118057720492160}. \end{aligned} \tag{33}$$

Proposition 7.8 (Band-edge regularity and curvature). *For $k = tn$ with $|n| = 1$,*

$$c_4(tn) - q = \frac{t^2}{4} \left[A \sum_i n_i^4 + B \sum_{i<j} n_i^2 n_j^2 \right] + O(t^4). \tag{34}$$

The directional coefficient ranges from

$$\kappa_{\text{diag}} = \frac{A+B}{12} = \frac{132329431693349}{3303982815494400}$$

on a body diagonal to

$$\kappa_{\text{axis}} = \frac{A}{4} = \frac{5}{48}$$

on a coordinate axis. Thus the Γ lift is quadratic but not the restriction of a single Hessian. At the upper edge,

$$c_4(R + \delta) = c_4(R) - \frac{132329431693349}{3303982815494400} |\delta|^2 + O(|\delta|^4), \tag{35}$$

so the R curvature is isotropic and negative.

Proof. Near Γ , $X_i = t^2 n_i^2 / 2 + O(t^4)$; insertion into (24) gives (34). Since $\sum_i n_i^4 \in [1/3, 1]$, $\sum_{i<j} n_i^2 n_j^2 = (1 - \sum_i n_i^4)/2$, and $A - B/2 > 0$, the stated extrema follow. If a Hessian existed at Γ , cubic symmetry would force it to be a scalar multiple of the identity, contradicting the two distinct directional coefficients. Near R , $X_i = 2 - \delta_i^2 / 2 + O(\delta_i^4)$; expanding the ratio in (24) gives (35). $\square$

7.2 Stable-rank $SU(N \geq 7)$ fourth-order reduction

Theorem 7.9 (Stable-rank local-invariant and denominator theorem). *For the complete connected fourth-order geometry, the maximum local tensor degree is six. For every integer $N \geq 7$, all local $SU(N)$ singlets are therefore exactly balanced in fundamental and antifundamental degree;*

quantity	count	quantity	count
connected supports	182440	candidate support/output pairs	895524
stable support/output classes	439	canonical ordered words	4171
exact-balance sign assignments	33500	charge-conjugation orbits	16750
unique balanced signatures	140	symbolic energy signatures	37500
resonant signatures	17073	nonresonant signatures	20427
distinct denominator polynomials	94	accidental roots, $N \geq 7$	0

Table 1: Exact stable-rank fourth-order geometry and denominator counts.

no independent determinant/epsilon sectors occur. The local Haar integrations reduce to the three balanced families $(1, 1)$, $(2, 2)$, and $(3, 3)$, with

$$\text{Wg}_1(e) = \frac{1}{N},$$

$$\text{Wg}_2(e) = \frac{1}{N^2 - 1}, \quad \text{Wg}_2((12)) = -\frac{1}{N(N^2 - 1)},$$

and

$$\text{Wg}_3(e) = \frac{N^2 - 2}{N(N^2 - 1)(N^2 - 4)},$$

$$\text{Wg}_3((12)) = -\frac{1}{(N^2 - 1)(N^2 - 4)}, \quad \text{Wg}_3((123)) = \frac{2}{N(N^2 - 1)(N^2 - 4)}.$$

The stable mixed-tensor irreducible channels are indexed by bipartitions (λ, μ) , with

$$C_2(\lambda, \mu) = \frac{N^2(|\lambda| + |\mu|) + N[\kappa(\lambda) + \kappa(\mu)] - (|\lambda| - |\mu|)^2}{2N}, \quad \kappa(\lambda) = \sum_i \lambda_i(\lambda_i + 1 - 2i).$$

Consequently every electric denominator is an explicit rational function of N . Exact enumeration yields the counts in Table 1; there are 94 distinct denominator polynomials and none has an accidental integer root for $N \geq 7$. Evaluating the stable bipartition engine at $N = 3$ agrees with the existing $SU(3)$ Dynkin-label engine on all 140 balanced token signatures.

Proof. A local invariant with n_f fundamental and $n_{\bar{f}}$ antifundamental indices requires $n_f - n_{\bar{f}} \equiv 0 \pmod{N}$. The verified local degree bound gives $|n_f - n_{\bar{f}}| \leq 6$; for $N \geq 7$ this congruence is equivalent to $n_f = n_{\bar{f}}$. The Weingarten coefficients are the exact inverses of the permutation Gram matrices through degree three. Stable mixed-tensor branching then gives the displayed bipartition Casimir. The geometry counts, denominator classification, absence of integer roots, and 140-signature regression are exact integer and symbolic gates in the certificate described in Appendix A. $\square$

Remark 7.10 (Remaining stable-rank contraction). *Theorem 7.9 completes the geometry, local Haar, channel, and denominator layers for $N \geq 7$. The remaining load-bearing step is to replace the $SU(3)$ -specific carrier/projector contractions in Stages 3C and 3G by symbolic walled-Brauer diagram tensors. The required final outputs are only q_N, A_N, B_N , together with the kernel gates in Proposition 7.7 and positivity of A_N, B_N . The ranks $N = 4, 5, 6$ require finite determinant-sector calculations; $SU(2)$ is separate because pseudoreality changes charge conjugation.*

8 Numerical summary and discussion

Table 2 collects the exact lower-order data. These entries are unchanged from the third-order certificate chain.

quantity	exact	decimal
C -odd flat branch, $O(y) / O(y^2) / O(y^3)$	$+1; +11/306; -109151/249696$	$1; 0.0359477; -0.4371356$
C -odd dispersive top, $O(y^2) / O(y^3)$	$41/306; -61751/249696$	$0.1339869; -0.2473048$
C -odd manifold width (y^2)	$5/51$	0.0980392
A_1^{++} bottom, $O(y) / O(y^2) / O(y^3)$	$-1; -217/1020; -54049/520200$	$-1; -0.2127451; -0.1039004$
E^{++} level, $O(y^2) / O(y^3)$	$223/1020; 52163/260100$	$0.2186275; 0.2005498$
C -even bandwidth (y^2)	$88/153$	0.5751634
C -even effective mass	$459/(44y^2)$	$10.43182/y^2$
per-bond ratio $ t_- / t_+ $	$5/22$	0.2272727

Table 2: Exact one-flux data through third order. The lowest C -odd branch is flat at every momentum through the displayed orders.

The new fourth-order data are summarized in Table 3.

quantity	exact	decimal
$c_4(\Gamma)$	$-20721577909065127111/7250590288602460800$	-2.857915988115
$c_4(X)$	$-17700498622147435111/7250590288602460800$	-2.441249321448
$c_4(M)$	$-4367164159624988707/1812647572150615200$	-2.409273720232
$c_4(R)$	$-3447362930970494909/1450118057720492160$	-2.377298119016
ΔW_4	$132329431693349/275331901291200$	0.480617869098
Γ body-diagonal coefficient	$132329431693349/3303982815494400$	0.040051489092
Γ axis coefficient	$5/48$	0.104166666667
R isotropic coefficient	$-132329431693349/3303982815494400$	-0.040051489092

Table 3: Exact projected $O(y^4)$ band data. The signs in the last row refer to the coefficient multiplying $|\delta|^2$ in the expansion about R .

Several consequences follow.

First, the Gauss-law mechanism is exact but order-limited. The elementary-cube CLS annihilate all link-mediated leakage through third order. Fourth-order site-mediated processes do not merely perturb this conclusion numerically: they produce the strictly positive form (23). The first nonzero bandwidth of the lowest T_1^{+-} branch is therefore

$$W(y) = \Delta W_4 y^4 + O(y^5).$$

The excitation is parametrically low-mobility at strong coupling, not all-orders immobile.

Second, the global band geometry is fixed without interval arithmetic. The minimum is at zero momentum and the maximum is at the Brillouin-zone corner. The rest-point degeneracy leaves a nonstandard signature: although $c_4(k) - c_4(\Gamma) = O(|k|^2)$, the coefficient depends on the approach direction. A conventional scalar effective mass at Γ is therefore not defined at this order. The upper edge at R has an ordinary isotropic negative curvature.

Third, the exact rest-frame series is

$$m_-(\Gamma, y) = \frac{8}{3} + y + \frac{11}{306}y^2 - \frac{109151}{249696}y^3 - \frac{20721577909065127111}{7250590288602460800}y^4 + O(y^5). \quad (36)$$

The magnitude of the fourth-order coefficient is already large, so naive polynomial extrapolation to moderate coupling is not controlled. A quantitative comparison with the continuum 1^{+-} mass additionally requires the string tension in the same Hamiltonian normalization.

Fourth, the quantum numbers remain useful for spectroscopy. The lowest C -odd one-flux excitation is T_1^{+-} , hence 1^{+-} -like at rest, not a scalar. The lower-order ordering $0^{++} < 2^{++} < 1^{+-}$ agrees qualitatively with continuum lattice spectroscopy [5, 6], but the present series does not predict the continuum mass.

Finally, the result provides both momentum- and real-space benchmarks for Hamiltonian lattice simulations and quantum hardware [13]. At small y , a correct implementation should reproduce zero dispersion through $O(y^3)$, followed by a bandwidth $\Delta W_4 y^4$, with the exact Γ , X , M , and R coefficients in Table 3. In real space, sandwiching the full 189-record kernel between cube-boundary maps must collapse to the 25-point stencil (28). This paired test is substantially more discriminating than a single rest-mass comparison.

9 Outlook

The fourth-order flatness and projected real-space geometry questions are now closed. Theorem 7.4 identifies the mobility operator as a local sum of squares of axial second differences and planar checkerboard differences. A remaining geometric refinement is to decompose the *unprojected* plaquette-space residual into boundary, co-boundary, and orthogonal sectors, clarifying which site-mediated processes mix the closed-surface subspace before projection.

The primary next direction is gauge-group dependence. The second-order part is closed: Theorem 5.2 proves positive signed hopping and a universal closed-surface flat branch for every $SU(N)$, $N \geq 3$. At fourth order, Theorem 7.9 closes the geometry, balanced Haar, representation channel, resonance, and denominator layers for the stable range $N \geq 7$. The single remaining stable-rank calculation is the symbolic walled-Brauer contraction replacing the $SU(3)$ -specific Stages 3C and 3G. It must produce q_N, A_N, B_N , pass the final kernel gates in Proposition 7.7, and prove $A_N, B_N > 0$ for all integers $N \geq 7$. The exceptional ranks $N = 4, 5, 6$ require finite rank-specific determinant-sector extensions, while $SU(2)$ requires a separate treatment because of pseudoreality and the resulting degeneration of charge conjugation.

For contact with continuum spectroscopy, the next high-value calculation is the Hamiltonian strong-coupling series for the string tension in the same normalization. Only the consistently expanded ratio $m_{1+-}(y)/\sqrt{\sigma(y)}$ can support a controlled comparison with the continuum lattice value. Higher mass orders, beginning with $O(y^5)$, are useful after that denominator is available; the fifth-order folded coefficient is known, but the full connected-geometry contraction remains a substantial computation.

Finally, the interaction of two or more compact closed-surface excitations is open. Flat-band systems are often dominated by interaction effects; here the relevant question is whether the lower-order incidence protection produces bound states, constrained kinetics, or new collective sectors before the single-particle $O(y^4)$ mobility becomes important.

A Verification chain

Every lower-order constant and structural claim retained from draft v0.1 is certified by the exact-arithmetic chain described there: `su3_moments_ext.py`, `su3_domino_d3.py`, `glueball_band_certificate_v2.py`, the two independent tromino checks, `cls_flat_band_certificate_v1_1.py`, and the document-level regression. That chain reported 387 hard gates and was rerun cold on June 13, 2026.

The complete from-scratch $SU(3)$ source chain has now been recovered and frozen from Stage

0 through Stages 1, 2, 3B, 3C, 3E, 3G, the folded des-Cloizeaux Stage 3I, and the final real-space assembly Stage 3J. Stage 3I is no longer a provenance gap; its recovered source has SHA-256

7662066c5516533ba531c414fc057c0f5c6e0d8bd1fe860a9c083fa8b2907abf.

Stage 3J records the folded-word source hash in the 189-record kernel metadata.

The fourth-order analytic update is certified independently by `y4_exact_analytic_factorization_certificate.py`. It reads the complete 189-record real-space kernel in the same cell/origin gauge as (22) and adds 26 exact gates. These verify:

1. the kernel record count, orientation ordering, and real-space Hermiticity;
2. $H_4(0) = qI_3$ and the exact value (21);
3. the 25-term Laurent support, inversion symmetry, and reduction to the $3 + 3 + 6$ cubic frequency orbits;
4. cancellation of the term linear in X_i and the exact coefficients

$$A = \frac{5}{12}, \quad B = \frac{17607806155349}{275331901291200};$$

5. the exact Γ , X , M , and R anchors;
6. the global-minimum and global-maximum hypotheses, bandwidth, and both band-edge curvature formulas.

The certificate status is **PASS**.

The real-space operator theorem is independently certified by `y4_exact_real_space_sos_certificate.py`. In exact Fraction arithmetic it forms $C^\dagger(H_4 - qI)C$ directly from all 189 kernel records, derives its 25 Laurent coefficients, and proves equality with (26). Its gates include exact Hermiticity, the scalar $H_4(0)$ value, the four stencil weights in (29), zero row sum, a finite-torus convolution regression, the $X/M/R$ generalized eigenvalues, and the exact bandwidth. The canonical semantic SHA-256 of the 189 sorted kernel records is

48a422a517c7c1e70b84fd88a0773943f81ae3f9bfafadbe2304f8eb7d2e9b77.

A separate single-block PyTorch script, `y4_sos_a100_validation.py`, applies both the full plaquette kernel and the 25-point scalar stencil on a periodic lattice and FFT-checks the numerator, Gram metric, generalized ratio, and parity-point lifts. This GPU run is an independent floating-point regression, not part of the exact proof.

The all-rank second-order extension is independently certified by `suN_closed_surface_banded_stage1_certificate.py`, with 34 exact symbolic gates. It verifies the four channel weights, the closed form and positivity of t_N , the $SU(3)$ anchors, the incidence-kernel flat eigenvalue, and the high-symmetry reconstruction identities.

The stable-rank fourth-order extension is certified separately by `y4_sun_stable_rank_stage1.py`. Starting from the exact 182440-support Stage-0 geometry, it recomputes every sign mask after canonicalization and proves the counts in Table 1. It verifies the balanced Weingarten families through degree three, 37500 symbolic energy signatures, 94 distinct denominator polynomials, absence of accidental integer roots for $N \geq 7$, and an exact 140/140 regression against the $SU(3)$ balanced-channel engine. A cold rerun regenerated the Stage-0 support file with SHA-256

5b3621cba76d4ac34b6ee89302399f07415db144735b07bbc926d0b370c4acdf

and reproduced every stable-rank gate. This certificate does not yet perform the symbolic walled-Brauer contractions needed for q_N, A_N, B_N .

The kernel SHA-256 frozen with the fourth-order $SU(3)$ certificate is

d2a4121a9798b2c364a52f7845fd7014ce2463563642470102cb080336a9fd51.

The earlier interval subdivision, which resolved 3248 boxes with no unresolved leaves, remains an independent numerical cross-check but is not needed for the analytic global theorem.

B Domino benchmark

For a minimal exact benchmark (two plaquettes sharing one link, seven links): through the order considered, the one-flux levels of (1) are, in y^2 units above $8/3 \mp y$,

$$C\text{-even: } \left\{ \frac{1769}{3060}, \frac{13}{20} \right\}, \quad C\text{-odd: } \left\{ \frac{31}{68}, \frac{17}{36} \right\}, \quad (37)$$

i.e. $\text{diag} \pm |t|$ with $(\text{diag}, |t|) = (\frac{1879}{3060}, \frac{11}{306})$ and $(\frac{71}{153}, \frac{5}{612})$ respectively. The upper C -even level sits exactly at the within-plaquette tower value $\frac{13}{20}$: in that combination the neighbor leakage cancels identically.

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